Preparing for Rugby World Cup

Daryl Collins - NZNOG 2019

Akamai
Experience the Edge

The Akamai Intelligent Platform

The world's largest on-demand, distributed computing platform delivers all forms of web content and applications

The Akamai Intelligent Platform:

240,000+	3,600+	1,700+	1300+	130
Servers	Locations	Networks	Cities	Countries

Typical daily traffic:

- More than **3 trillion** requests served
- Delivering over **50 Terabits/second**

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Akamai CDN

- Akamai CDN is comprised of distinct, geographically & topologically disparate clusters
- We believe having lots of clusters in lots of places gives us better performance than a few large sites
- Each cluster in NZ is effectively an island

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How Akamai CDN Works

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How Akamai CDN Works

Akamai is a DNS-based CDN

Users querying a DNS-based CDN will be returned different A (and AAAA) records for the same hostname depending on the resolver the request comes from

This is called “mapping”

The better the mapping, the better the CDN

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How Akamai CDN Works

When content is requested from Akamai, multiple criteria is examined to choose the optimal server:

- Content Availability
- Latency & Packet Loss
- Cluster resources - CPU load, memory, and storage space
- Network utilisation

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How Akamai CDN Works

Example of Akamai mapping:

```
[Auckland]% host www.example.com
www.example.com   CNAME a888.w10.akamai.net.
e5211.b.akamaiedge.net.  A       219.88.206.18
e5211.b.akamaiedge.net.  A       219.88.206.83

[Christchurch]% host www.example.com
www.example.com   CNAME a888.w10.akamai.net.
e5211.b.akamaiedge.net.  A       219.88.204.32
e5211.b.akamaiedge.net.  A       219.88.204.25
```


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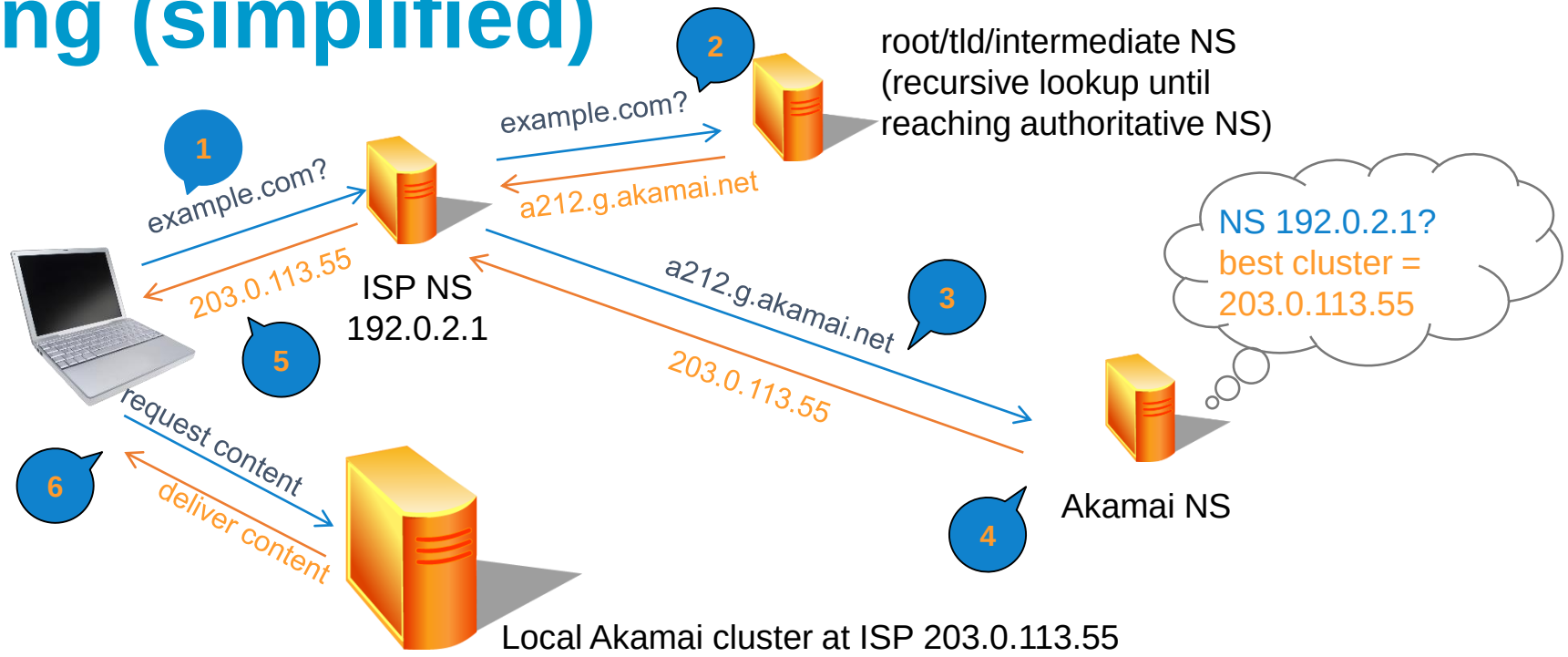
Mapping (simplified)



1. End-user requests www.example.com from ISP NS
2. ISP NS recursively looks up www.example.com and is referred to authoritative Akamai NS
3. ISP NS asks authoritative Akamai NS
4. Akamai NS looks up the IP of requestor (ISP NS) and replies with IP of optimal cluster to serve content
5. ISP NS replies to end-user with IP of optimal cluster
6. End-user requests content from the optimal cluster

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Akamai in New Zealand

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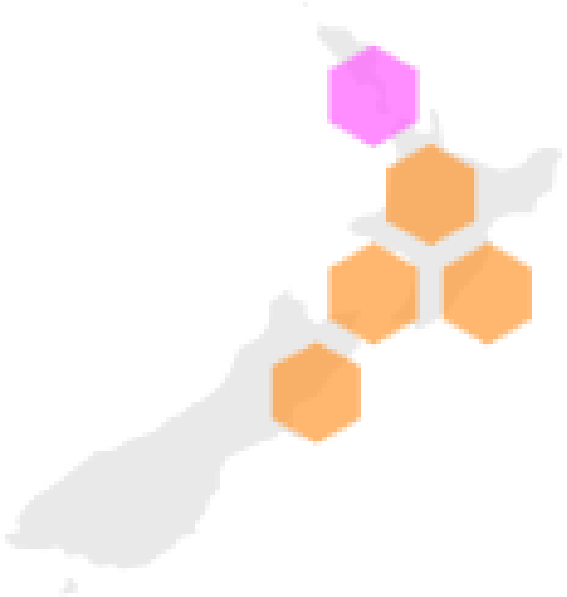
Akamai's CDN in New Zealand

- Inside ISP networks
  - Primarily in Auckland, Wellington, Christchurch
- Internet Exchanges in Auckland
  - AKL IX, Megaport IX, APE



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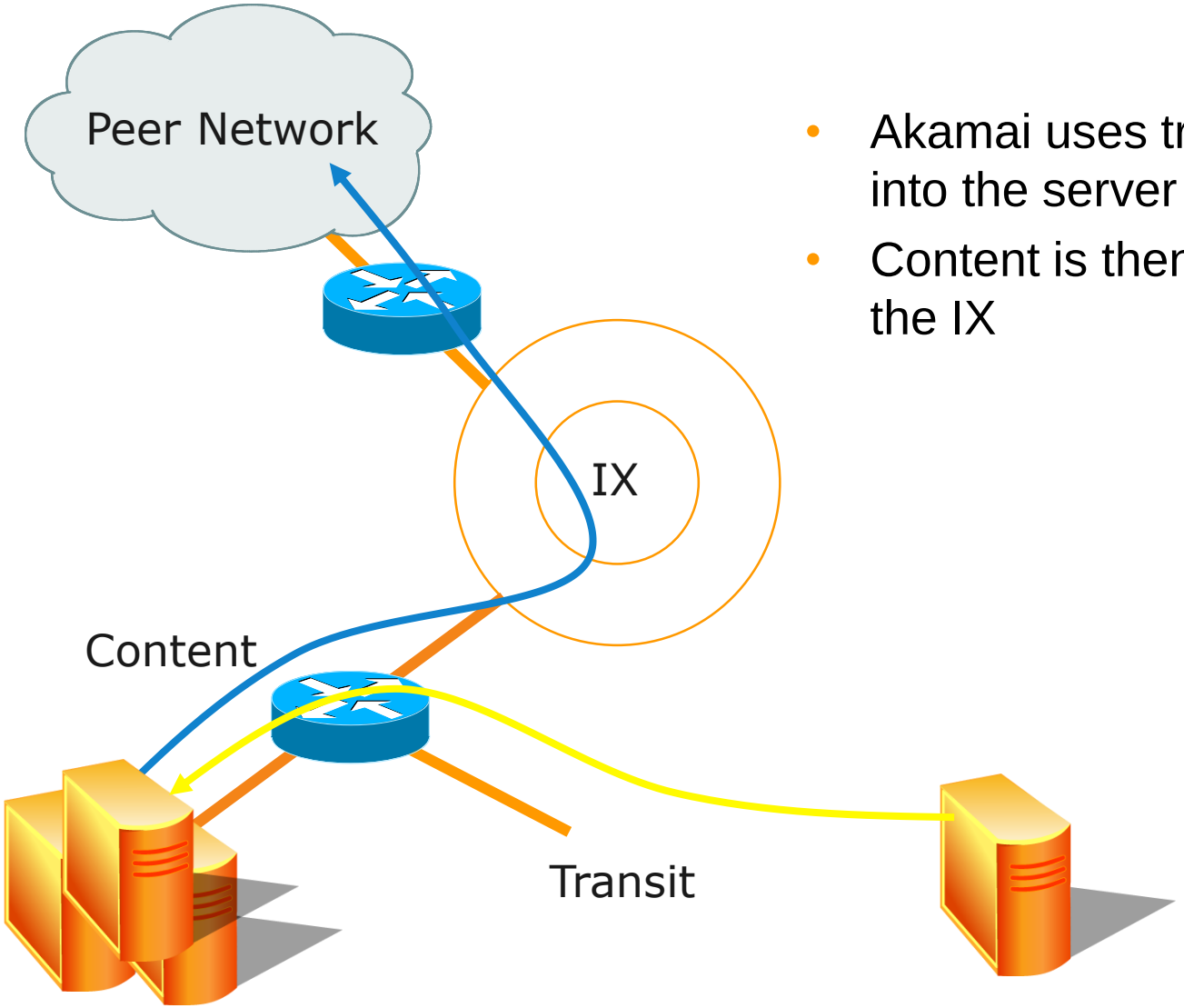
How Akamai uses Internet Exchanges



- Akamai uses transit to pull content into the server
- Content is then served to peers over the IX

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How Akamai uses Internet Exchanges

Akamai typically does not announce large blocks of address space because no single location has a large number of servers

- It is not uncommon to see a single /24 from Akamai at an IX

This does not mean you won’t see a lot of traffic

- How many servers does it take to fill 10 gigabits these days?

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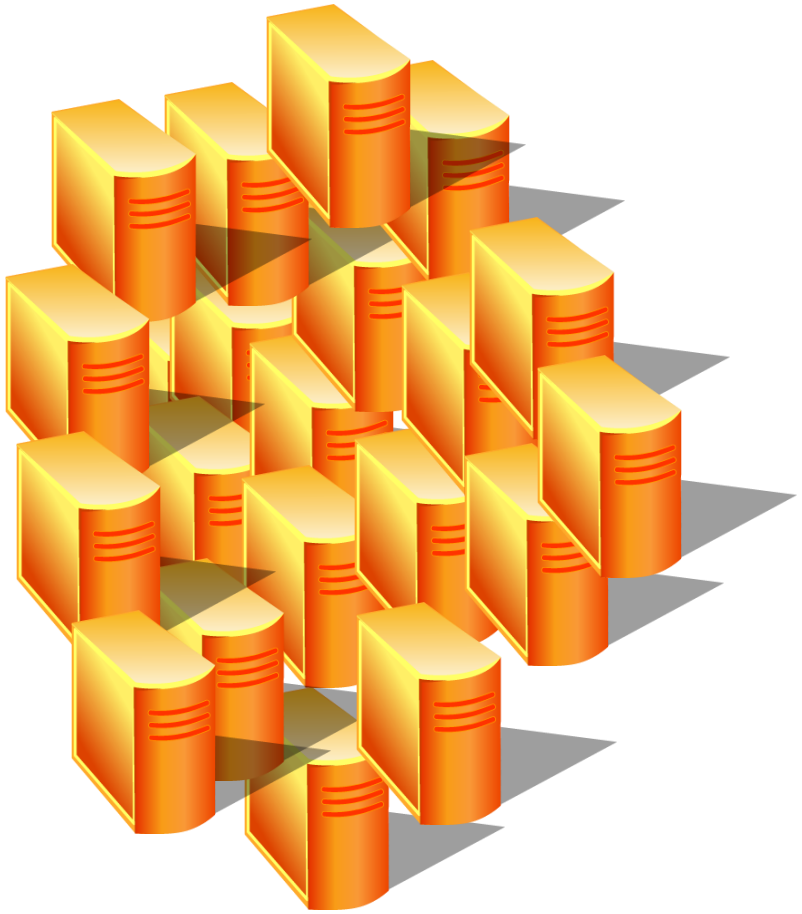
Why don’t I get all the Akamai content via Peering?



- No single cluster can accommodate all Akamai content
- Clusters get more efficient with with size
- Some content requires specialized servers only present in Infrastructure clusters
- Some content is only present in specific geographies
- Akamai prefers on-net clusters over peering

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How much traffic will I see?

- Event traffic generally ties in with marketing. Are your end-users being marketed to?
- Are the matches at times when your end-users are on your network?
- Are the devices end-users likely to use connected to your network?
- Multiply out your market share by the event size

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Where will the traffic come from?

Generally traffic will come from similar locations to what you see today.

- One way to get an indication is to see where a1.w16.akamai.net resolves to at different times of day.

How does Akamai handle congested paths?

- If high latency or packet loss to your DNS resolver is detected we will attempt to serve you from a cluster that does not see this latency/loss.

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Akamai Deployments

- Akamai is working with existing Akamai Accelerated Network Partners to add capacity
  - Upgrade existing clusters
  - Additional clusters in existing locations
  - Additional clusters in new locations

- Akamai is adding capacity to clusters connected to Internet Exchanges

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Peering

- Peering maximises the locations event traffic can be sourced from
  - Establish bi-lateral BGP sessions with Akamai at all common IX's
- Akamai Peering details can be found on PeeringDB (can yours?)
  - https://as20940.peeringdb.com/
  - peering@akamai.com

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DNS resolvers

Think about the DNS resolvers your end-users use.

- Google or OpenDNS resolvers
  - Akamai will map based on the subnet that the end-user is located in due to EDNS0 Client Subnet.
- Other DNS resolvers (including ISP resolvers)
  - Akamai maps based on the external IP address of the DNS resolver.

Tip: Announce your routes consistently. Advertise DNS prefixes and end-user prefixes together

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Checking Akamai's view of your DNS request

```
dig +short TXT whoami.ds.akahelp.net
"ns" "2001:db8::abcd"
"ecs" "203.0.113.0/24/0"
"ip" "203.0.113.132"
```

- ns - is the unicast IP address of the requesting recursive resolver
- ecs - is the ECS client subnet included in the query
- ip - is a representative IP address selected at random by Akamai's authoritative nameserver from the ECS client subnet.

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Questions?

Contacts

- Peering Sessions/Mapping Queries
  - peering@akamai.com
- Cluster Upgrades
  - Dave Dickford - ddickfor@akamai.com
- Other questions?
  - Daryl Collins - daryl@akamai.com

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